

Course: CPSC 8430: Deep Learning Homework 2

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Title: Video Caption Generation

Repository:

Find the project repository here:

<https://github.com/VASPARITHARUNKUMAR/CPSC-8430---Deep-Learning---HW2>

Problem Overview:

The task is to generate descriptive captions for input videos using a sequence-to-sequence (seq2seq) model. The input consists of video data, while the output is a stream of text that narrates the video's actions. The process employs Recurrent Neural Networks (RNNs) to achieve this.

Requirements:

- Python
- Torch
- Cuda (GPU)
- Scipy
- Pickle
- Pandas
- Numpy

Dataset:

The model has been trained and tested using the dataset supplied by the professor. It consists of 1,450 videos for training and 100 videos for testing. Additionally, we have .json files for both training and testing, which contain label information needed to train the model.

Model Design:

The model employs a sequence-to-sequence architecture comprising an encoder and a decoder, both of which use Gated Recurrent Units (GRUs). GRUs are chosen over LSTMs for their computational efficiency due to fewer parameters, making them suitable for the relatively short sequences in the dataset.

- **Encoder:** Encodes video frames into a format that can be processed by the decoder.
- **Decoder:** Generates captions based on encoded information and attention mechanisms.

The model also incorporates several key components:

1. **Attention Layer:** Enables the model to concentrate on specific portions of the video at each time step, enhancing the accuracy of caption generation.

2. **Schedule Sampling:** Alleviates exposure bias by replacing some ground truth tokens with model-generated predictions during training.

Hyperparameters:

The model was trained using the following parameters:

- **Epochs:** 100
- **Batch Size:** 128
- **Teacher Learning Ratio:** 0.7
- **Dropout Rate:** 0.3
- **Optimizer:** Adam
- **Hidden Layer Size:** 512
- **Learning Rate:** 0.0001

Performance Evaluation:

Upon completion of training for 100 epochs, the model attained a **BLEU score** of **0.7121041558090666** and a final **epoch loss** of **1.824320912361145**, reflecting strong performance in accurately generating captions for the test videos. The training metrics were collected on the Palmetto cluster with the following architecture.

Training the data: python compute.py

Testing the data:

```
python test_func.py $'/home/tvaspar/DL_HW2/MLDS_hw2_1_data/testing_data/feat'  
$'test_data.txt'
```

Bleu Score: sh hw2_seq2seq.sh

Tokenization and Vocabulary:

The model employs special tokens to manage sequences effectively:

- **<PAD>**: Used to pad sequences to a uniform length
- **<BOS>**: Signals the start of a sentence
- **<EOS>**: Marks the sentence's conclusion
- **<UNK>**: Denotes unknown or out-of-vocabulary words

A vocabulary was created, filtering out words that appeared fewer than four times in the dataset, resulting in a dictionary with 6,443 unique tokens.

Conclusion:

Through sequential modeling with GRUs and attention mechanisms, the model successfully generates video captions with a high BLEU score of **0.7121041558090666**. This project demonstrates the effectiveness of sequence-to-sequence models in processing video data and highlights the importance of attention and schedule sampling in improving model robustness.